

- QUESTION 1 If $z = 1 + i\sqrt{3}$ express z in the following form 2
- $$z = r(\cos\theta + i\sin\theta)$$
- and hence plot on the Argand diagram:
- i) $\bar{z}$ 1
- ii) $\frac{12}{z}$ 1
- iii) z^2 1
- iv) $\sqrt{z}$ 1
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- QUESTION 2 Solve for x and y : 3
- $$(x + iy)(2 + 3i) = 18i - 1$$
-
- QUESTION 3 Sketch on the Argand diagram where 3
- $$|z| < 4 \quad \text{and} \quad \frac{-\pi}{3} \leq \arg z \leq \frac{\pi}{4} \quad \text{both hold.}$$
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- QUESTION 4 Prove that $(6 + i\sqrt{108})^{12}$ is real. 2
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- QUESTION 5 Find the modulus of the product of the roots of the equation: 2
- $$(2 + 3i)z^2 + (5 - 2i)z + (18 + i) = 0$$
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- QUESTION 6 Find $\sqrt{-8 - 6i}$ and hence solve the equation: 4
- $$z^2 - (5 - i)z + 8 - i = 0$$

- QUESTION 7 P, Q and R represent the complex numbers 4
 z_1, z_2 and z_3 respectively on the Argand diagram.
 Given that $z_1 - z_2 = i(z_2 - z_3)$ show this information
 on a diagram and hence fully describe ΔPQR .
- QUESTION 8 The point P on the Argand diagram represents the complex 3
 number z where z satisfies the equation:

$$|z - 2i| = \operatorname{Im}(z + 2i)$$

 Find the Cartesian equation of the locus of P and give a
 geometric description of the locus of P.
- QUESTION 9 By applying De Moivre's Theorem, and by also expanding 4
 $(\cos\theta + i\sin\theta)^4$ express $\tan 4\theta$ as a polynomial in terms of $\tan\theta$.
- QUESTION 10 i) If ω is a complex solution of the equation $z^3 = 1$ show that ω^2 2
 is also a solution of the equation and that:

$$1 + \omega + \omega^2 = 0$$

 (ii) Given that ω is also a root of $z^3 + bz^2 + cz - 12 = 0$, 2
 find the values of b and c given they are both real.

END OF EXAMINATION

Question 1

$$z = 1 + i\sqrt{3}$$

$$= 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

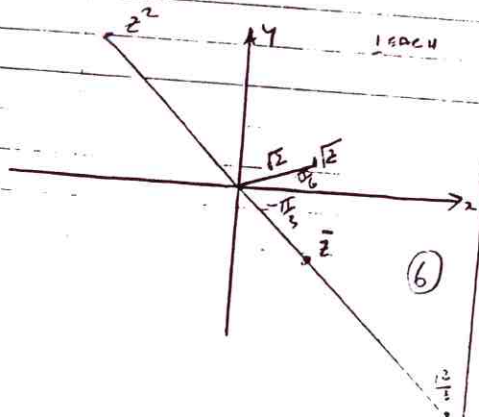
$$= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\bar{z} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\frac{z}{2} = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$$

$$z^2 = 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$\sqrt{z} = \sqrt{2}\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$



(6)

Question 2

$$(x+iy)(2+3i) = 18i - 1$$

$$(2x-3y) + i(2y+3x) = 18i - 1$$

$$\therefore 2x-3y = -1 \quad (1)$$

$$2y+3x = 18 \quad (2)$$

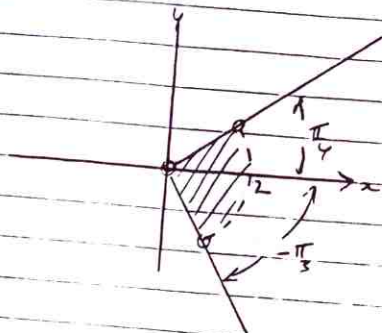
$$) \times 2 \Rightarrow 4x-6y = -2 \quad (3)$$

$$) \times 3 \Rightarrow 6y+9x = 54 \quad (4)$$

$$+ (4) \Rightarrow 13x = 52 \quad (3)$$

$$x = 4$$

Question 3



$\frac{1}{2} \times \pi \times 12^2$

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Question 4

$$(6+i\sqrt{108})^{12} = (6+6\sqrt{3}i)^{12}$$

$$= 12^{12} \left(\frac{6+6\sqrt{3}i}{12}\right)^{12}$$

$$= 12^{12} \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{12}$$

$$= 12^{12} \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{12}$$

$$= 12^{12} \left(\cos 4\pi + i\sin 4\pi\right)$$

$$= 12^{12} (\cos 0 + i\sin 0)$$

$$= 12^{12}$$

WHICH IS REAL

(2)

Question 5

$$(2+3i)z^2 + (5-2i)z + (18+i) = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(5-2i) \pm \sqrt{(5-2i)^2 - 4(2+3i)(18+i)}}{2(2+3i)}$$

$$= \frac{-(5-2i) \pm \sqrt{25-20i-4(36+33i+54+32i)}}{2(2+3i)}$$

$$= \frac{-(5-2i) \pm \sqrt{25-20i-4(88+65i)}}{2(2+3i)}$$

$$= \frac{-(5-2i) \pm \sqrt{25-20i-352-260i}}{2(2+3i)}$$

$$= \frac{-(5-2i) \pm \sqrt{-327-280i}}{2(2+3i)}$$

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Question 6

$$\text{Let } z = x+iy$$

$$\therefore x+iy = \sqrt{-8-6i}$$

$$\therefore x^2-y^2+2xyi = -8-6i$$

$$\therefore x^2-y^2 = -8$$

$$2xy = -6$$

$$\text{mult } y = \frac{-3}{x}$$

$$\therefore x^2 - \frac{9}{x^2} = -8$$

$$\therefore x^4 + 8x^2 - 9 = 0$$

$$(x^2+9)(x^2-1) = 0$$

$$\therefore x = \pm 1, y = \pm 3$$

$$\therefore z = \pm(1-3i)$$

$$3^2 - (5-i)^2 - 4(5-i) = 0$$

$$3 = \frac{5-i \pm \sqrt{(5-i)^2 - 4(5-i)}}{2}$$

$$= \frac{5-i \pm \sqrt{25-10i-1-32+4i}}{2}$$

$$= \frac{5-i \pm \sqrt{-8-6i}}{2}$$

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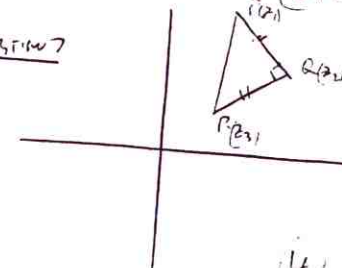
$$= \frac{5-i \pm \sqrt{-8-6i}}{2}$$

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Question 7



Q7 (cont) $\triangle PQR$ IS AN ISOSCELES
RIGHT ANGLED TRIANGLE
WITH RT $\angle$ AT Q (z_2).

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Question 8

$$|z - zi| = \ln(z + zi)$$

Let $z = x + iy$

$$|(x + iy - zi)| = \ln(x + iy + zi)$$

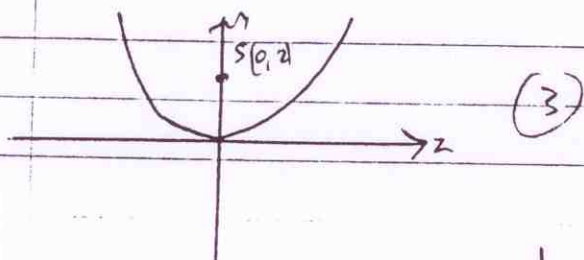
$$|x + i(y - z)| = y + z \quad \perp$$

$$\therefore \sqrt{x^2 + (y - z)^2} = y + z$$

$$x^2 + y^2 - 4y + 4 = y^2 + 4y + 4$$

$$x^2 = 8y \quad \perp$$

$$\therefore x^2 = 4 \cdot 2y$$



A Parabola Vertex O and focus (0,2)

Question 9

$$(\cos 4\theta + i \sin 4\theta) = (\cos \theta + i \sin \theta)^4 \text{ Let } z = \cos \theta, i = \sin \theta$$

$$\cos 4\theta + i \sin 4\theta = C^4 + 4C^3 i + 6C^2 i^2 + 4C i^3 + i^4$$

$$= C^4 + 4C^3 i - 6C^2 + -4C i + 1$$

$$\cos 4\theta = C^4 - 6C^2 + 1 \quad \sin 4\theta = 4C^3 - 4C$$

$$\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4C^3 - 4C}{C^4 - 6C^2 + 1} \quad \perp$$

Dividing by C^4

$$\therefore \tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{4\frac{C^3}{C^4} - 4\frac{C}{C^4}}{1 - 6\frac{C^2}{C^4} + \frac{1}{C^4}} \quad (4)$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$$

Question 10

1) $z^3 = 1$

$$\therefore (z-1)(z^2+z+1)$$

$$(3-1)(3^2+3+1)=0$$

if w is a complex root & then eqn

$$w \neq 1 \quad \perp$$

$$\therefore w \text{ must satisfy } z^2 + z + 1 = 0$$

$$\therefore w^2 + w + 1 = 0 \quad \perp$$

OR if w is a root

$$\text{then } w^3 = 1$$

$$\text{but } (w^2)^3 = (w^3)^2 = 1 \quad \perp$$

$$\therefore w^2 \text{ is also a root}$$

$$\text{now } z = 1$$

$$\therefore 3 \text{ roots are } 1, w, w^2$$

$$\text{now } \Sigma d \text{ for } z^3 - 1 = 0$$

$$\Sigma d = -\frac{b}{a} = 0 \quad \perp$$

$$1 + w + w^2 = 0$$

ii) if w is a root so is w^2

$$\text{now } \alpha \beta \gamma = -\frac{d}{a} = 12$$

$$w - w^2 \cdot y = 12$$

$$y = 12$$

$$\Sigma d = 12 + w + w^2 = -b$$

$$\therefore 11 + (1 + w + w^2) = -b \quad \perp$$

$$\therefore b = -11 \quad *$$

$$\alpha \beta + \beta \gamma + \gamma \alpha = w^3 + w^2 y + w y = C$$

$$= 1 + y(w + w^2)$$

$$= 1 + 12(w + w^2) = C$$

$$= 1 - 11(1 + w + w^2) - 12 = C$$

$$\therefore C = -11 \quad * \quad \perp$$